

The revised manuscript proposes AQ-ROM, an integrated framework that combines exploratory analysis, Lasso-based feature selection, XGBoost regression, and Bayesian optimization to predict temperature and rainfall from air-quality and meteorological variables. The authors have made a visible effort to improve the methodological description, clarify that separate XGBoost models are trained for the two targets, document the Bayesian optimization space, discuss rainfall-specific characteristics, and add an ablation analysis.

Major issues that require correction -

- Table 3 has been mentioned twice, both in the ablation studies and statistical-validation table.
- The statistical-validation table (Table 3) contains placeholders such as “Mean $\pm$ SD” and “Actual p-value” instead of numerical values.
- The manuscript states that the preprocessed dataset was “split randomly” into 80% training and 20% testing data. However, the response to Reviewer 3 states that temporal order was preserved and that the oldest observations were used for training and the newest for testing. These are materially different protocols. There should be consistency in the manuscript and response to the reviewers.
- At one place the manuscript has reported 7,000 observations, still it uses “N observations from M cities” in the abstract and dataset description.
- The reviewers requested standard baselines such as standalone XGBoost, Linear Regression, Random Forest, LightGBM, CatBoost and LSTM. The response acknowledges the limitation but postpones most comparisons to future work. This does not fully answer the reviewer’s experimental concern.
- There is a narrative discussion of XGBoost feature importance, but the manuscript should provide the actual importance values or an explainability figure for each target. A general statement that pollutants are important is insufficient for a paper whose title emphasizes interpretability.

Minor issues -

- Table 1 has been described as containing standard deviation. But the displayed table does not include a standard-deviation column.
- Proofreading is required for expressions like “The is the input”, etc.